EXP NO: 3 HANDLING MISSING AND INAPPROPRIATE DATA IN A DATASET

AIM:

Demonstrate an experiment to handle missing data and inappropriate data in a Data set using Python Pandas Library for Data Preprocessing.

DATASET:

CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill
1	20-25	4	Ibis	veg	1300
2	30-35	5	LemonTree	Non-Veg	2000
3	25-30	6	RedFox	Veg	1322
4	20-25	-1	LemonTree	Veg	1234
5	35+	3	Ibis	Vegetarian	989
6	35+	3	Ibys	Non-Veg	1909
7	35+	4	RedFox	Vegetarian	1000
8	20-25	7	LemonTree	Veg	2999
9	25-30	2	Ibis	Non-Veg	3456
9	25-30	2	Ibis	Non-Veg	3456
10	30-35	5	RedFox	non-Veg	-6755

CODE:

```
import pandas as pd
import numpy as np
df = pd.read_excel("Hotel_Dataset.xlsx")
print("Original Dataset:")
print(df.head())
print("\nDataset Information:")
df.info()
print("\nMissing Values:")
print(df.isnull().sum())
print("\nDuplicate Records:")
print(df.duplicated().sum())
df.drop_duplicates(inplace=True)
numeric_columns = df.select_dtypes(include=np.number).columns
for column in numeric_columns:
 df[column] = df[column].fillna(df[column].median())
categorical_columns = df.select_dtypes(include="object").columns
for column in categorical_columns:
```

```
 if df[column].isnull().any():
    df[column] = df[column].fillna(df[column].mode()[0])
for column in numeric_columns:
 Q1 = df[column].quantile(0.25)
 Q3 = df[column].quantile(0.75)
 IQR = Q3 - Q1
 lower_limit = Q1 - 1.5 * IQR
 upper_limit = Q3 + 1.5 * IQR
 df[column] = df[column].clip(lower_limit, upper_limit)
print("\nMissing Values After Preprocessing:")
print(df.isnull().sum())
print("\nDuplicate Records After Preprocessing:")
print(df.duplicated().sum())
print("\nPreprocessed Dataset:")
print(df.head())
df.to_excel("Preprocessed_Hotel_Dataset.xlsx", index=False)
print("\nData preprocessing completed successfully.")
```

OUTPUT:

```
Original Dataset:
 CustomerID Age_Group Rating(1-5) Hotel FoodPreference Bill NoOfPax EstimatedSalary
Age_Group.1
0 1 20-25 4 Ibis veg 1300 2 40000 20-25
1 2 30-35 5 LemonTree Non-Veg 2000 3 59000 30-35
2 3 25-30 6 RedFox Veg 1322 2 30000 25-30
3 4 20-25 -1 LemonTree Veg 1234 2 120000 20-25
4 5 35+ 3 Ibis Vegetarian 989 2 45000 35+
Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
 # Column Non-Null Count Dtype
--- ------ -------------- -----
 0 CustomerID 11 non-null int64
 1 Age_Group 11 non-null str
 2 Rating(1-5) 11 non-null int64
 3 Hotel 11 non-null str
 4 FoodPreference 11 non-null str
 5 Bill 11 non-null int64
 6 NoOfPax 11 non-null int64
 7 EstimatedSalary 11 non-null int64
 8 Age_Group.1 11 non-null str
dtypes: int64(5), str(4)
```

memory usage: 924.0 bytes
Missing Values:
CustomerID 0
Age_Group 0
Rating(1-5) 0
Hotel 0
FoodPreference 0
Bill 0
NoOfPax 0
EstimatedSalary 0
Age_Group.1 0
dtype: int64
Duplicate Records:
1
/Users/yuva/Desktop/Code_Hub/FDS_Exp/Exp_3/exp.py:25: Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. This behavior is deprecated and will be removed in a future version. Explicitly pass'str' to `include` to select them, or to `exclude` to remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-selectdtypes for details on how to writecode that works with pandas 2 and 3.
 categorical_columns = df.select_dtypes(include="object").columns
Missing Values After Preprocessing:
CustomerID 0
Age_Group 0
Rating(1-5) 0
Hotel 0
FoodPreference 0
Bill 0
NoOfPax 0
EstimatedSalary 0
Age_Group.1 0
dtype: int64
Duplicate Records After Preprocessing:
0
Preprocessed Dataset:
 CustomerID Age_Group Rating(1-5) Hotel FoodPreference Bill NoOfPax EstimatedSalary Age_Group.1
0 1 20-25 4 Ibis veg 1300.0 2.0 40000.0 20-25
1 2 30-35 5 LemonTree Non-Veg 2000.0 3.0 59000.0 30-35
2 3 25-30 6 RedFox Veg 1322.0 2.0 30000.0 25-30
3 4 20-25 0 LemonTree Veg 1234.0 2.0 120000.0 20-25
4 5 35+ 3 Ibis Vegetarian 989.0 2.0 45000.0 35+
Data preprocessing completed successfully

RESULT:

Thus the code is executed successfully.